

7. The finiteness theorem for the invariants of finite groups

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The finiteness theorem for the invariants of finite groups.

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In what follows an entirely elementary proof of finiteness for the invariants of finite groups is to be given — one based only on the theory of symmetric functions — and at the same time it gives an actual specification of the full system; whereas the usual proof, based on Hilbert's theorem on module bases (Ann. 36), is only an existence proof.^{*)}

Let the finite group $\mathfrak{H}$ consist of the h linear transformations (with non-vanishing determinant) $A_1, \dots, A_h$, where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^n a_{1\nu}^{(k)} x_\nu, \dots, x_n^{(k)} = \sum_{\nu=1}^n a_{n\nu}^{(k)} x_\nu,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. The group $\mathfrak{H}$ therefore carries the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. Since the identity must be contained among $A_1, \dots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$. By an integral rational (absolute) invariant of the group is meant an integral rational function of $x_1, \dots, x_n$ which remains identically unchanged under the application of $A_1, \dots, A_h$; for such an invariant $f(x)$ one therefore has

$$f(x) = f(x^{(1)}) = \dots = f(x^{(h)}) = \frac{1}{h} \sum_{k=1}^h f(x^{(k)}). \quad (1)$$

1.

Formula (1) says that $f(x)$ is an integral rational symmetric function of the rows of quantities $(x^{(1)}), \dots, (x^{(h)})$ — and indeed, since each summand $f(x^{(k)})$ contains only the one row $(x^{(k)})$, this is the simplest case, usually called the *one-form* case. By the known theorem on symmetric functions of rows of quantities,^{*)} $f(x)$ is therefore integrally and rationally representable by the elementary symmetric functions of these rows, that is, by the coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the “Galois resolvent”

$$\begin{aligned} \Phi(z, u) &= \prod_{k=1}^h (z + u_1 x_1^{(k)} + \dots + u_n x_n^{(k)}) \\ &= z^h + \sum G_{\alpha\alpha_1\dots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \dots u_n^{\alpha_n}, \end{aligned} \quad \left(\begin{array}{c} \alpha + \alpha_1 + \dots + \alpha_n = h \\ \alpha \neq h \end{array} \right),$$

where the $G_{\alpha\alpha_1\dots\alpha_n}(x)$ are invariants of degree $\alpha_1 + \dots + \alpha_n$ in the x 's. Thus it is proved:

The coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the Galois resolvent form a full system of invariants of the group, in such a way that every invariant can be represented integrally and rationally by these finitely many invariants.

2.

Starting from (1), one may also make the following still more elementary consideration,^{**)} which at the same time leads to a second full system. Put

$$f(x) = a + b x_1^{\mu_1} \dots x_n^{\mu_n} + \dots + c x_1^{\nu_1} \dots x_n^{\nu_n},$$

where $a, b, \dots, c$ denote constants. Then by (1) one has

$$\begin{aligned} h \cdot f(x) &= h \cdot a + b \sum_{k=1}^h (x_1^{(k)})^{\mu_1} \dots (x_n^{(k)})^{\mu_n} + \dots + c \sum_{k=1}^h (x_1^{(k)})^{\nu_1} \dots (x_n^{(k)})^{\nu_n} \\ &= h \cdot a + b \cdot J_{\mu_1\dots\mu_n} + \dots + c \cdot J_{\nu_1\dots\nu_n}. \end{aligned}$$

^{*)} Cf., for example, Weber, *Lehrbuch der Algebra* (2nd ed.), vol. 2, § 57.

^{*)} Cf. the note to 2.

^{**) This amounts to a proof of the theorem on symmetric functions of rows of quantities in the above-mentioned one-form case.}

Every invariant can therefore be composed integrally and linearly from the special invariants

$$J_{\mu_1 \dots \mu_n} = \sum_{k=1}^h (x_1^{(k)})^{\mu_1} \dots (x_n^{(k)})^{\mu_n},$$

and it is enough to prove the assertion for these special ones. But, apart from a numerical factor, $J_{\mu_1 \dots \mu_n}$ is the coefficient of $u_1^{\mu_1} \dots u_n^{\mu_n}$ in the expression

$$S_\mu = \sum_{k=1}^h (u_1 x_1^{(k)} + \dots + u_n x_n^{(k)})^\mu,$$

where $\mu = \mu_1 + \dots + \mu_n$, and this expression is the μ -th power sum of the h linear forms

$$\xi_1 = u_1 x_1^{(1)} + \dots + u_n x_n^{(1)}, \dots, \xi_h = u_1 x_1^{(h)} + \dots + u_n x_n^{(h)}.$$

It is known that the infinitely many power sums S_μ are integral rational combinations of

$$S_1, S_2, \dots, S_h,$$

whose coefficients are given by the invariants

$$J_{\mu_1 \dots \mu_n} \quad (\mu_1 + \dots + \mu_n \leq h).$$

Thus a second full system has been obtained:

A full system of invariants of the group is given by all invariants $J_{\mu_1 \dots \mu_n}$ for which $\mu_1 + \dots + \mu_n \leq h$, where h denotes the order of the group.

The connection with 1. is established by observing that the power sums $S_1, \dots, S_h$ can be replaced by the elementary symmetric functions of $\xi_1, \dots, \xi_h$, which leads to the full system of the $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ given there. Both results show that all invariants can be expressed integrally and rationally by those whose degree in the x 's does not exceed the order h of the group.

3.

Consequences for rational representation can be drawn from what has just been proved. Every rational absolute invariant can be represented — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally by the coefficients $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent $\Phi(z, u)$. This theorem can also be proved easily in various ways without passing through the finiteness theorem; it is already formulated in Weber II, § 58.*)

4.

Finally, it should be mentioned that the results derived here are implicitly contained in my paper “Fields and systems of rational functions”;*) more precisely, the full system given in 1. is contained in Theorems VI and VII, and a proof of rational representation independent of this finiteness theorem is contained in Theorem

*)The proof given there, however, is not sound. It is only shown that the function $\Psi(t)$ in formula (7) has invariants as coefficients, but not that these invariants are rationally expressible by the coefficients of $\Phi(t)$. This gap is avoided if, in representing the $x_i^{(k)}$ by ξ_k , one uses a known differentiation procedure instead of Lagrange's interpolation formula. This is the relation obtained by differentiating the identity $\Phi(-\xi_k, u) = 0$ with respect to all u :

$$\left[\frac{\partial \Phi}{\partial u_i} - x_i^{(k)} \frac{\partial \Phi}{\partial z} \right]_{z=-\xi_k} = 0.$$

Accordingly, for each rational function $\omega(x)$ the following formula must replace formulas (7) and (8) in Weber:

$$\omega(x_1^{(k)}, \dots, x_n^{(k)}) = \omega \left(\frac{\partial \Phi / \partial u_1}{\partial \Phi / \partial z}, \dots, \frac{\partial \Phi / \partial u_n}{\partial \Phi / \partial z} \right) \Big|_{z=-\xi_k},$$

which contains only coefficients of $\Phi(z, u)$, and whose summation over all k , for invariants $\omega(x)$, gives the desired representation.

*)Math. Ann. 76, p. 161 (1915).

III.**) In the present special case, however, the course of the proof and the result simplify considerably in comparison with the general theory. I was led to apply the general investigations to the invariants of finite groups by conversations with E. Fischer, from whom, in substance, the proof given under 2 — in the general case the full system that appears there is not rationally definable — and the note to 3 also originate.

Erlangen, May 1915.

**) For relative invariants, Theorems VIII and IX contain a proof of finiteness which is likewise based on a different foundation from the usual one, but is also only an existence proof; the reason is that the relative invariants do not form a field.